

Discrete-Time Signals and Systems

Signals and Systems

- **Signals:** A signal is defined as any physical quantity that varies with time, space, or any other independent variable or variables. By a signal we mean any variable that carries or contains some kind of information that can, for example, be conveyed, displayed or manipulated.
 - speech, which we encounter for example in telephony, radio and everyday life,
 - biomedical signals, such as electrocardiogram and electroencephalogram (brain signals),
 - sound and music, such as reproduced by the compact disc player,
 - video and image, which most people watch on the television, and
 - radar signals, which are used to determine the range and bearing of distant targets.

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 - radar signals, which are used to determine the range and bearing of distant targets.
- **Systems:** Signals may be processed further by systems, which may modify them or extract additional information from them. A system is an entity that processes a set of input signals to yield another set of output signals.

Signals Processing

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- **Digital Signal Processing:** DSP is concerned with the digital representations of signals and the use of digital processors to analyze, modify, or extract information from signals.
- **Advantages of DSP:**
 - Guaranteed accuracy
 - Flexibility
 - Superior performance
 - Adaptability
 - Perfect reproducibility
 - Multiplexing
 - Data logging

Classification of Signals

- **Continuous-Time and Discrete-Time Signals:** A continuous-time or analog signal, $x(t)$, is defined for a continuum of values of the independent variable time t . A discrete-time signal, $x(n)$, is defined only at discrete-time n .

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- **Continuous-Valued and Discrete-Valued Signals:** The value of a continuous-time or discrete-time signal can be continuous or discrete. If a signal takes on all possible values on a finite or an infinite range, it is said to be continuous-valued signal. Alternatively, if the signal takes on values from a finite set of possible values, it is said to be a discrete-time signal.

Classification of Signals

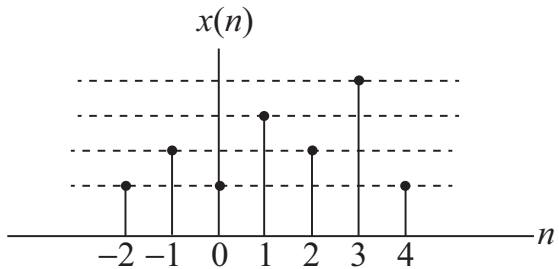
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- **Multichannel and Multidimensional Signals:** Multichannel signals are generated by multiple sources or multiple sensors. A signal is called *M-dimensional* if its value is a function of M independent variables.

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- **Deterministic and Random Signals:** Any signal that can be described by an explicit mathematical relationship is called deterministic. The nature and amplitude of such a signal at any time can be predicted. A signal whose values cannot be predicted precisely but are known only in terms of probabilistic description, such as mean value or mean-square value, is a random signal.

Basic Operations on Discrete-Time Signals

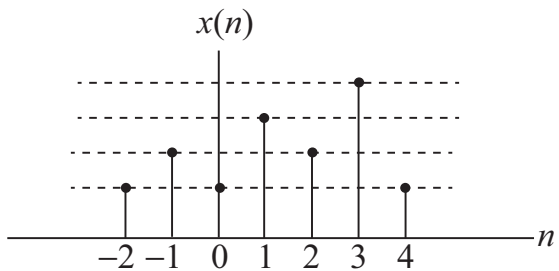
- **Discrete-Time Signal or Sequence:** A discrete-time signal $x(n]$ is defined only for integer values of n and is undefined for noninteger values of n .



$$x(n) = \{\cdots, \quad x(-3), \quad x(-2), \quad x(-1), \quad \underset{\uparrow}{x(0)}, \quad x(1), \quad x(2), \quad x(3), \quad x(4), \quad \cdots\}$$

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- **Basic Operations on Discrete-Time Signals:** Let $w(n)$, $x(n)$ and $y(n)$ be the DT signal and α is a scalar quantity.

- Signal Addition Operation: $y(n) = w(n) + x(n)$
- Scalar Addition Operation: $y(n) = \alpha + x(n)$
- Signal Multiplication Operation: $y(n) = w(n)x(n)$
- Scalar Multiplication Operation: $y(n) = \alpha x(n)$

Example

● **Example:** Given the following two signals of length 5 defined for $-1 \leq n \leq 3$:

$$w(n) = \{1.5, \underset{\uparrow}{2}, 3.4, -5, 10\}$$

$$x(n) = \{2.2, \underset{\uparrow}{3}, 2, 4.2, 8\}$$

Determine the following new signals: (a) $y_1(n) = w(n) + x(n)$, (b) $y_2(n) = 3 + x(n)$, (c) $y_3(n) = w(n)x(n)$, (d) $y_4(n) = \frac{3}{2}x(n)$.

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● **Solution: (a)** Adding the sample values of two signals $w(n)$ and $x(n)$ to form a new signal $y_1(n)$.

$$y_1(n) = w(n) + x(n)$$

$$y_1(-1) = w(-1) + x(-1) = 1.5 + 2.2 = 3.7$$

$$y_1(0) = w(0) + x(0) = 2 + 3 = 5$$

$$y_1(1) = w(1) + x(1) = 3.4 + 2 = 5.4$$

$$y_1(2) = w(2) + x(2) = -5 + 4.2 = -0.8$$

$$y_1(3) = w(3) + x(3) = 10 + 8 = 18$$

Example contd...

therefore, $y_1(n) = \{3.7, \underset{\uparrow}{5}, 5.4, -0.8, 18\}.$

(b)

$$y_2(n) = 3 + x(n)$$

$$\begin{aligned} y_2(n) &= \{3 + 2.2, \quad 3 + \underset{\uparrow}{3}, \quad 3 + 2, \quad 3 + 4.2, \quad 3 + 8\} \\ &= \{5.2, \quad \underset{\uparrow}{6}, \quad 5, \quad 7.2, \quad 11\} \end{aligned}$$

(c)

$$y_3(n) = w(n)x(n)$$

$$y_3(-1) = w(-1)x(-1) = 1.5 \times 2.2 = 3.3$$

$$y_3(0) = w(0)x(0) = 2 \times 3 = 6$$

$$y_3(1) = w(1)x(1) = 3.4 \times 2 = 6.8$$

$$y_3(2) = w(2)x(2) = -5 \times 4.2 = -21$$

$$y_3(3) = w(3)x(3) = 10 \times 8 = 80$$

Example contd...

therefore,

$$y_3(n) = \{3.3, \underset{\uparrow}{6}, 6.8, -21, 80\}$$

(d) Multiplying a scalar value $\frac{3}{2}$ to each sample of a signal $x(n)$ to form a new signal $y_4(n)$.

$$y_4(n) = \frac{3}{2}x(n)$$

$$\begin{aligned} y_4(n) &= \left\{2.2 \times \frac{3}{2}, \underset{\uparrow}{3} \times \frac{3}{2}, 2 \times \frac{3}{2}, 4.2 \times \frac{3}{2}, 8 \times \frac{3}{2}\right\} \\ &= \{3.3, \underset{\uparrow}{4.5}, 3, 6.3, 12\} \end{aligned}$$

Transformations of the Independent Variable

● **Transformations of the Independent Variable (Time):** Three types of transformations.

- Time-shifting
- Time-scaling (Decimation and Interpolation)
- Time-reversal or folding

Transformations of the Independent Variable

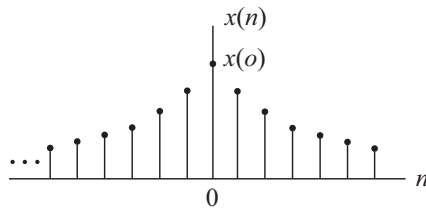
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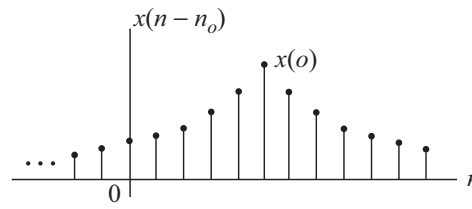
● **Time-shifting:**

● **Time Delay (Shift-right):** $x(n) \longrightarrow x(n - n_0)$

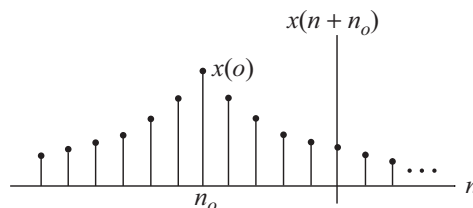
● **Time Advance (Shift-left):** $x(n) \longrightarrow x(n + n_0)$



(a)



(b)

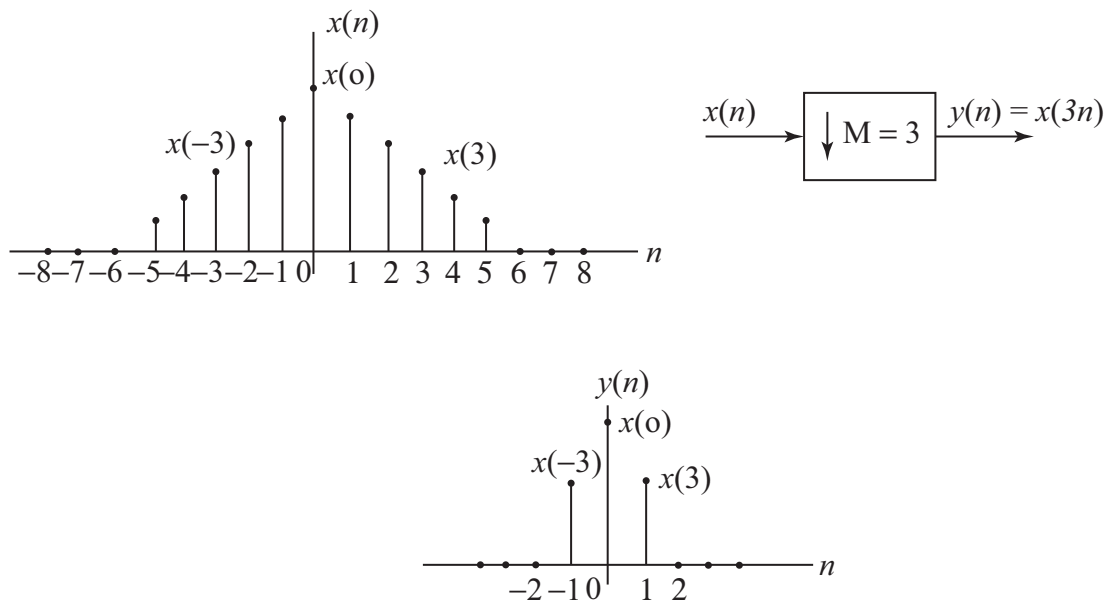


(c)

Transformations of the Independent Variable contd...

- **Time-scaling (Decimation and Interpolation):**
 - **Decimation (Downsampling):** Downsampling by a factor of M . Discarding $(M - 1)$ in-between samples.

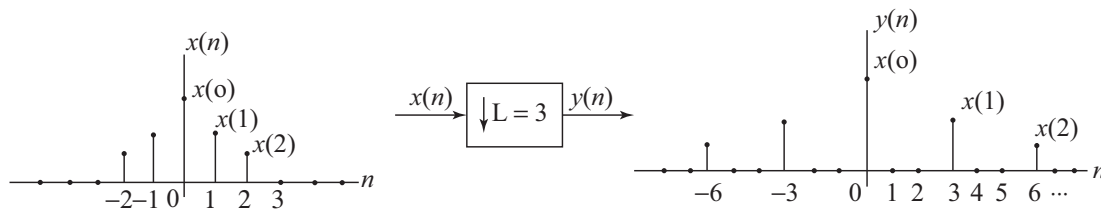
$$y(n) = x(Mn)$$



Transformations of the Independent Variable contd...

- **Interpolation (Upsampling):** Upsampling by a factor-of L . Inserting $L - 1$ zero-valued samples between each two consecutive samples of $x(n)$.

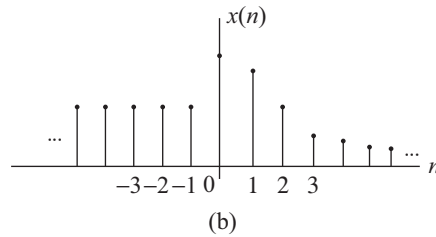
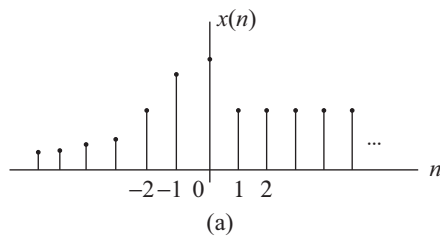
$$y(n) = \begin{cases} x\left(\frac{n}{L}\right) & n = 0, \pm L, \pm 2L, \dots \\ 0 & \text{otherwise} \end{cases}$$



Transformations of the Independent Variable contd...

- 🟢 **Time-reversal or folding:** Reflection of the signal about the time origin $n = 0$.

$$x(n) \longrightarrow x(-n)$$



Some Basic Discrete-time Signals

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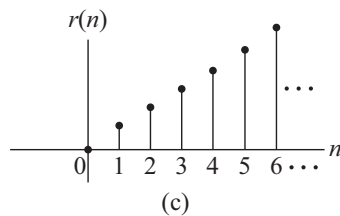
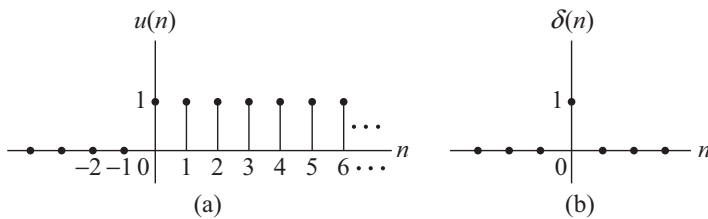
● **Unit Impulse Signal (or Unit Sample Signal):** $\delta(n) = \begin{cases} 1 & n = 0 \\ 0 & n \neq 0 \end{cases}$

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● **Unit Ramp Signal:** $r(n) = nu(n) = \begin{cases} n & n \geq 0 \\ 0 & n < 0 \end{cases}$



Periodic Signals

● **Periodic and Aperiodic Signals:** A discrete-time signal $x(n)$ is periodic if and only if

$$x(n) = x(n + N) \quad \text{for all } n$$

where N is a positive integer. A periodic signal $x(n)$ remains unchanged when time-shifted by one period.

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- **Periodicity of Discrete-Time Sinusoidal Signals:** The periodicity of discrete-time sinusoidal signals are characterized by the following properties:

- **Discrete-time sinusoidal signals are not always periodic in time:** *A discrete-time sinusoidal signal is periodic only if its frequency is a rational number.*

$$\cos(\omega n) = \cos(2\pi f_d n)$$

$$= \cos[2\pi f_d (n + N)]$$

$$\cos(\omega n) = \cos[2\pi f_d n + 2\pi f_d N]$$

Periodic Signals contd...

The two sides are equal provided

$$f_d N = k$$

$$f_d = \frac{k}{N} \quad \text{rational number (ratio of integers)}$$

Discrete-time sinusoidal signals are always periodic in frequency: *Discrete-time sinusoidal signals with frequencies separated by an integer multiple of 2π are identical.*

$$x(n) = \cos(\omega n) = \cos[(\omega \pm 2\pi m)n] = \cos[\omega n \pm 2\pi nm]$$

Highest oscillation rate in a discrete-time sinusoidal signal occurs at $\omega = \pi$ or $f_d = \frac{1}{2}$: *The highest rate of oscillation in a discrete-time sinusoidal signal is attained when $\omega = \pi$ or ($\omega = -\pi$) or, equivalently, $f_d = \frac{1}{2}$ or ($f_d = -\frac{1}{2}$).*

Energy and Power Signals

- **Energy and Power Signals:** The energy of a discrete-time signal is given by

$$E_x = \sum_{n=-\infty}^{\infty} |x(n)|^2.$$

The power of a discrete-time signal is given by

$$P_x = \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N |x(n)|^2.$$

- $0 < E_x < \infty$, i.e., if E_x is finite and $P_x = 0$, then $x(n)$ is referred to as an *energy signal*.
- If $E_x = \infty$, but $0 < P_x < \infty$, i.e., P_x is finite, then $x(n)$ is referred to as a *power signal*.
- All finite length signals are energy signals.
- All finite periodic signals are power signals; however not all power signals are periodic e.g. unit step signal.

Even and Odd Signals

● **Even and Odd Signals:** Let $x(n)$ be a real-valued discrete-time signal.

● **Even Signal:** $x(n)$ is even if it is identical with its reflection about the origin, i.e.,

$$x(n) = x(-n) \quad \text{for all } n$$

● An even signal is symmetrical about the vertical axis.

● **Odd Signal:** $x(n)$ is odd if

$$x(n) = -x(-n) \quad \text{for all } n$$

● An odd signal is antisymmetric about the vertical axis. An odd signal must necessarily be 0 at $n = 0$.

Discrete-Time Systems and Properties

- **Discrete-Time Systems:** A discrete-time system is a system in which discrete-time input signals are transformed into discrete-time output signals.

$$x(n) \longrightarrow y(n) \quad \text{or} \quad y(n) = \mathsf{T}[x(n)] \quad \text{or} \quad x(n) \longrightarrow \boxed{\mathsf{T}[\cdot]} \longrightarrow y(n)$$

where $\mathsf{T}[\cdot]$ is a system operator.

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- **System Properties:**
 - Linear or nonlinear
 - Time-invariant or time- varying
 - Memoryless or with memory
 - Causal or noncausal
 - Stable or unstable.

Systems Properties contd...

● **Linear and Nonlinear Systems:** A system is linear if it follows the superposition principle, that is, if it obeys the properties of additivity and homogeneity.

● Additivity property: $x_1(n) + x_2(n) \rightarrow y_1(n) + y_2(n)$.

● Homogeneity property: $ax_1(n) \rightarrow ay_1(n)$, where a is an arbitrary constant.

● These two properties can be combined into a single statement as

$$ax_1(n) + bx_2(n) \rightarrow ay_1(n) + by_2(n)$$

Example: For the following input-output relationships, determine whether the system is linear.

$$y(n) = nx(n)$$

Solution: Consider three arbitrary inputs $x_1(n)$, $x_2(n)$, and $x_3(n)$.

$$x_1(n) \longrightarrow y_1(n) = nx_1(n)$$

$$x_2(n) \longrightarrow y_2(n) = nx_2(n)$$

$$x_3(n) \longrightarrow y_3(n) = nx_3(n)$$

System Properties contd...

Let $x_3(n)$ be a linear combination of $x_1(n)$ and $x_2(n)$. That is,

$$x_3(n) = ax_1(n) + bx_2(n)$$

where a and b are arbitrary scalars. If the system is linear, then

$$y_3(n) = ay_1(n) + by_2(n)$$

Consider the LHS of the above equation

$$\begin{aligned} y_3(n) &= nx_3(n) \\ &= n[ax_1(n) + bx_2(n)] \\ &= anx_1(n) + bnx_2(n) = ay_1(n) + by_2(n) = RHS \end{aligned}$$

Since $LHS = RHS$, we conclude that the system is linear.

System Properties contd...

● **Time-Varying and Time-Invariant Systems (or Shift-Invariant Systems):** A system is time invariant if the behavior and characteristics of the system are fixed over time, Or

- A system is said to be time invariant if a time shift in the input signal causes an identical time shift in the output signal.
- For example, if $y(n)$ is the output of a discrete-time system corresponding to the input $x(n)$, a time-invariant system will have $y(n - n_o)$ as the output when $x(n - n_o)$ is the input, i.e.,

$$\text{if } x(n) \longrightarrow y(n) \quad \text{then} \quad x(n - n_o) \longrightarrow y(n - n_o)$$

Example: For the following input-output relationships, determine whether the system is time-invariant.

$$y(n) = nx(n)$$

Solution: Let $y_1(n)$ and $y_2(n)$ be the outputs corresponding to the inputs $x_1(n)$, and $x_2(n)$ respectively. Then

System Properties contd...

$$y_1(n) = nx_1(n)$$

$$y_2(n) = nx_2(n)$$

Let $x_2(n) = x_1(n - n_o)$, If the system is time-invariant, then

$$y_2(n) = y_1(n - n_o)$$

Consider the LHS of the above equation, we obtain

$$y_2(n) = nx_2(n) = nx_1(n - n_o)$$

Now, consider

$$y_1(n) = nx_1(n)$$

$$y_1(n - n_o) = (n - n_o)x_1(n - n_o)$$

Clearly, $y_2(n) \neq y_1(n - n_o)$, and hence, this system is not time-invariant.

System Properties contd...

- **Causal Systems:** A system is causal if the present value of the output depends only on the present and past values of the input signal, or
 - A system is causal or nonanticipatory (also known as physically realizable), if the output at any time n_o depends only on values of the input at the present time and in the past i.e., for $n \leq n_o$.
 - Mathematically, if

$$x_1(n) = x_2(n) \quad n \leq n_o$$

and the system is causal, then

$$y_1(n) = y_2(n) \quad n \leq n_o$$

- *All memoryless systems are causal systems*, since the output responds only to the current value of the input.

Example: For the following input-output relationships, determine whether the system is causal.

$$y(n) = nx(n)$$

System Properties contd...

Solution: Consider the output $y(n)$ at $n = n_o$,

$$y(n)|_{n=n_o} = y(n_o) = nx(n)|_{n=n_o} = n_o x(n_o)$$

Now, consider the output $y(n)$ at $n = -n_o$,

$$y(n)|_{n=-n_o} = y(-n_o) = nx(n)|_{n=-n_o} = -n_o x(-n_o)$$

In both the cases the present output depends upon the present input, hence, the system is causal.

System Properties contd...

● **Stable Systems:** A system is said to be *bounded-input, bounded-output* (BIBO) stable if and only if every bounded input results in a bounded output.

● The output of a stable system does not diverge if the input does not diverge.

● Signal $x(n)$ is said to be bounded if its magnitude does not grow without bound, i.e.,

$$|x(n)| \leq B_x < \infty \quad \text{for all } n$$

● A system is BIBO stable if, for any bounded input $x(n)$, the response $y(n)$ is also bounded. That is,

$$|x(n)| \leq B_x < \infty \quad \text{implies} \quad |y(n)| \leq B_y < \infty$$

Example: For the following input-output relationships, determine whether the system is stable.

$$y(n) = nx(n)$$

System Properties contd...

Solution: Assume that $x(n)$ is bounded, that is,

$$|x(n)| \leq B_x < \infty \quad \text{for all } n$$

Using the given input-output relation

$$y(n) = nx(n),$$

we can write

$$|y(n)| = |nx(n)| = |n||x(n)| = |n|B_x$$

As $n \rightarrow \infty$, the output $|y(n)| \rightarrow \infty$, which is unbounded, hence, the system is unstable.

System Properties contd...

- **Systems with and Without Memory:** A system is said to possess *memory* (with memory system or dynamic system) if its output signal depends on past or future values of the input signal.

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- **Systems with and Without Memory:** A system is said to possess *memory* (with memory system or dynamic system) if its output signal depends on past or future values of the input signal.
- In contrast, a system is said to be *memoryless* (or static or instantaneous) if its output signal depends only on the present value of the input signal.
 - For example the system specified by the relationship

$$y(n) = 2x(n) + 3x^2(n)$$

is memoryless, as the value of $y(n)$ at any particular time n_o depends only on the value of $x(n)$ at that time.

- An example of a discrete-time system with memory is an *accumulator* or *summer*

$$y(n) = \sum_{k=-\infty}^n x(k)$$

and a second example is a delay

$$y(n) = x(n - 1)$$

System Properties contd...

- **Invertibility and Inverse Systems:** A system is invertible if by observing the output, we can determine its input.

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- **Invertibility and Inverse Systems:** A system is invertible if by observing the output, we can determine its input.
- we can construct an inverse system that when cascaded with the given system yields an output equal to the original input to the given system.
 - For a system to have an inverse (or be invertible) distinct inputs must lead to distinct outputs (one-to-one mapping between an input and the corresponding output).
 - If a system produces an identical output for two different inputs, it does not have an inverse.
 - For example, a rectifier, specified by an equation $y(n) = |x(n)|$ is noninvertible because the rectification operation can not be undone.
 - The system described by

$$y(n) = \cos[x(n)]$$

is not invertible because different inputs (such as $x(n) + 2k\pi$, $k = 0, \pm 1, \pm 2, \dots$) yield an identical output.

Discrete-Time Signals and Systems

Thank You!